

Module 3 In-class Coding Practice - Polymorphism

 Published

Complete the following skeleton codes:

Method Overriding Skeleton

```
class Animal {  
    // Method: makeSound  
    // - Implement this method to make a sound  
}  
  
class Dog extends Animal {  
    // Override the makeSound method to make a specific sound  
}  
  
class Cat extends Animal {  
    // Override the makeSound method to make a specific sound  
}  
  
public class PolymorphismExample {  
    public static void main(String[] args) {  
        // Create instances of Dog and Cat  
        // Use the makeSound method to demonstrate polymorphism  
    }  
}
```

Interface Polymorphism Skeleton

```
interface Shape {  
    double area();  
}  
  
class Circle implements Shape {  
    // Implement the area method for a circle  
}  
  
class Square implements Shape {  
    // Implement the area method for a square  
}  
  
public class PolymorphismExample {  
    public static void main(String[] args) {  
        // Create instances of Circle and Square  
        // Use the area method to demonstrate polymorphism  
    }  
}
```

Grading Rubric

Example 1: Method Overriding (Maximum 50 Points)

Criteria	Points
Class and Method Definition	
- Class names match the entity (e.g., Animal, Dog, Cat)	5
- Method makeSound defined correctly in each class with proper overriding	10
Implementation and Output	
- Methods produce the correct output when called	20
- Polymorphism demonstrated by using Animal references to Dog and Cat objects	10
Code Structure	
- Code is well-structured with proper indentation and formatting	5
- Proper use of access modifiers (public, override)	5
Documentation	
- Clear comments in classes and methods to describe their purpose	5
- Code follows naming conventions for variables, methods, and classes	5
Total Points (Example 1)	50

Example 2: Interface Polymorphism (Maximum 50 Points)

Criteria	Points
Interface and Class Definition	
- Shape interface correctly defines the area method	5
- Class names match the entities (e.g., Circle, Square)	5
- Properly implement the area method in Circle and Square	15
Implementation and Output	
- Methods produce the correct output when called	15
- Polymorphism demonstrated by using Shape references to Circle and Square objects	15
Code Structure	
- Code is well-structured with proper indentation and formatting	5
- Proper use of access modifiers (public, override)	5
Documentation	
- Clear comments in classes and methods to describe their purpose	5
- Code follows naming conventions for variables, methods, and classes	5
Total Points (Example 2)	50

Points 100

Submitting a file upload

Due	For	Available from	Until
Sep 28	Everyone	-	Oct 22 at 11:59pm